

Introduction to Markov Decision Processes and Dynamic Programming in Reinforcement Learning

1 - Consider the gridworld and the policy shown for it in Figure 3-5 of the book. Write a Python program that obtains the optimal state-value function values shown in the figure and displays them as a 5×5 matrix with a precision of one decimal place.

Hint: For this purpose, you may use the solution of the linear equations given by Equation 3-14 in the book, or use the equation

$$V = (I - \gamma P)^{-1} R$$

Just note that P and R in this equation correspond to a Markov reward process. You may enter the values of these two variables manually in the code, or obtain them from the variables that exist in a Markov decision process. If you have another solution in mind, that is also fine; you may use it and explain it in your report.

2 - Complete Exercise 3-15 of the book.

Next, consider the environment of Example 4-1 in the book and perform the following exercises for it.

3 - Using the policy evaluation algorithm, evaluate the policy in which each possible move is selected with equal probability, and display its values. Take the discount factor to be 0.9.

4 - Using the policy iteration algorithm with a discount factor of 0.9 and starting from a deterministic policy, find and display the optimal policy and the values of the optimal state-value function. (Try to display the policy graphically, for example, using the symbols ($<$, $\wedge$, $>$, $\vee$).)

5 - Using the value iteration algorithm with a discount factor of 0.9, find the optimal policy according to the algorithm in the book. Did you observe any difference in execution between this algorithm and the policy iteration algorithm? How many iterations of this algorithm, in finding the optimal state-value function, were sufficient to obtain the optimal policy?

For the implementations, use Python. Do not display decimal values with more than three decimal places. (Always build your programs incrementally, from small to large, and once you are confident that one part is correct, add another part to it.)

Submit your answers to the above questions and the results of your experiments in a PDF report together with your programs.

Submit a short video of your own execution of the program, together with an explanation of the experiments you performed.

Please note the following:

- Assignments may be submitted up to two days after the deadline. For each day of delay, 10% of the grade will be deducted.
- Avoid colloquial writing and spelling mistakes, as they will result in a deduction of marks.
- Do not include images of handwritten material in your report; no credit will be given for them.
- Only assignments submitted through the submission system will be graded. Under no circumstances should you email your assignment.
- No credit will be given for copied or downloaded assignments.

Best wishes

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